

Solving by completing the square

ANSWER KEY • FOR TEACHERS



Interactive version

REMEMBER

Squares

Level 1 • threshold

g8.expand

1.

Which expression is equal to $(x + 3)^2$?

☐ A $x^2 + 9$

☒ B $x^2 + 6x + 9$ —
 $(x + 3)^2 = x^2 + 2 \cdot$
 $x \cdot 3 + 3^2 = x^2 +$
 $6x + 9.$

☐ C $x^2 + 6x + 6$

☐ D $x^2 + 3x + 9$

Level 1 • threshold

g9.qe.2

2.

Solve $(x + 5)^2 = 4$.

Answer: -3; -7

Level 1 • threshold

g8.complete-sq

3.

Which number in the box makes $x^2 + 10x + \square$ the square of a binomial?

☒ A 25 — $x^2 + 10x +$
 $25 = (x + 5)^2,$
 because $2 \cdot x \cdot 5 =$
 $10x.$

☐ B 5

☐ C 10

☐ D 100

WORKED EXAMPLES

Worked examples

YOUR TURN

Level 2 • satisfactory

g9.qe.4

Solve $x^2 - 12x + 20 = 0$ by completing the square.

WORKED SOLUTION

$x^2 - 12x = -20$, so $x^2 - 12x + 36 = 16$, that is, $(x - 6)^2 = 16$. Then $x - 6 = 4$ or $x - 6 = -4$: $x = 10$ or $x = 2$.

Answer: 10; 2

YOUR TURN

Level 2 · satisfactory

g9.qe.4

In lesson 1.1, Ona tried to solve $x(x - 3) = 4$ and got it wrong. Solve it properly: multiply out, then complete the square.

WORKED SOLUTION

$x^2 - 3x = 4$, so $x^2 - 3x + \frac{9}{4} = \frac{25}{4}$, that is, $(x - \frac{3}{2})^2 = \frac{25}{4}$. Then $x - \frac{3}{2} = \frac{5}{2}$ or $x - \frac{3}{2} = -\frac{5}{2}$: $x = 4$ or $x = -1$. The roots Ona could not find are 4 and -1 .

Answer: 4; -1

YOUR TURN

Level 2 · satisfactory

g9.qe.4

Solve $9x^2 - 18x + 5 = 0$ by completing the square.

WORKED SOLUTION

Dividing by 9: $x^2 - 2x + \frac{5}{9} = 0$, so $x^2 - 2x = -\frac{5}{9}$. Adding 1: $(x - 1)^2 = \frac{4}{9}$. Then $x - 1 = \frac{2}{3}$ or $x - 1 = -\frac{2}{3}$: $x = \frac{5}{3}$ or $x = \frac{1}{3}$.

Answer: $x_1 = \frac{1}{3}$, $x_2 = \frac{5}{3}$

CHECK

Check yourself

Level 1 · threshold

g9.qe.3

1.

Which number must you add to $x^2 - 8x$ to make it the square of a binomial?

Answer: 16

Level 1 · threshold

g9.qe.4

2.

Which equation is the same as $x^2 - 6x = 16$?

- ☒ A $(x - 3)^2 = 25$ — ☐ B $(x - 3)^2 = 16$ ☐ C $(x + 3)^2 = 25$ ☐ D $(x - 6)^2 = 52$
- Add $(-3)^2 = 9$ to both sides: $x^2 - 6x + 9 = 25$.

Level 2 · satisfactory

g9.qe.3

3.

Write $x^2 + 7x + 10$ in the form $(x + p)^2 + q$. Enter p and q , in this order, separated by a semicolon.

WORKED SOLUTION

$x^2 + 7x + 10 = x^2 + 7x + \frac{49}{4} - \frac{49}{4} + 10 = (x + \frac{7}{2})^2 - \frac{9}{4}$, because $10 = \frac{40}{4}$. So $p = \frac{7}{2}$ and $q = -\frac{9}{4}$.

Answer: $p = \frac{7}{2}$, $q = -\frac{9}{4}$

4.

Solve $x^2 + 3x - 10 = 0$ by completing the square.

WORKED SOLUTION

$x^2 + 3x = 10$, so $x^2 + 3x + \frac{9}{4} = \frac{49}{4}$, that is, $(x + \frac{3}{2})^2 = \frac{49}{4}$. Then $x + \frac{3}{2} = \frac{7}{2}$ or $x + \frac{3}{2} = -\frac{7}{2}$: $x = 2$ or $x = -5$.

Answer: 2; -5

5.

Complete the square in each equation. Which one has **exactly one** root?

- ☐ A $x^2 + 14x + 40 = 0$
 ☐ B $x^2 + 49 = 0$
 ☒ C $x^2 + 14x + 49 = 0$
 ☐ D $x^2 + 14x + 50 = 0$
- It is $(x + 7)^2 = 0$
 , so $x + 7 = 0$: one
 root, $x = -7$.

6.

Put the steps for solving $x^2 + 4x - 12 = 0$ in the right order.

- 1 $x^2 + 4x = 12$
- 2 $x^2 + 4x + 4 = 16$
- 3 $(x + 2)^2 = 16$
- 4 $x + 2 = 4$ or $x + 2 = -4$
- 5 $x = 2$ or $x = -6$

7.

Solve $3x^2 + 12x + 6 = 0$ by completing the square. Show each step and give the exact roots.

MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

- | | |
|---|------|
| Divides by 3 and moves the constant: $x^2 + 4x = -2$. | 1 pt |
| Completes the square correctly: $(x + 2)^2 = 2$. | 1 pt |
| Both roots: $x = -2 + \sqrt{2}$ and $x = -2 - \sqrt{2}$. | 1 pt |

WORKED SOLUTION

$3x^2 + 12x + 6 = 0$. Divide by 3: $x^2 + 4x + 2 = 0$, so $x^2 + 4x = -2$.

Add $2^2 = 4$ to both sides: $x^2 + 4x + 4 = 2$, that is, $(x + 2)^2 = 2$.

Since $2 > 0$: $x + 2 = \sqrt{2}$ or $x + 2 = -\sqrt{2}$.

Answer: $x = -2 \pm \sqrt{2}$, that is, $x = -2 + \sqrt{2}$ or $x = -2 - \sqrt{2}$.

EXAM TASK

Not quite a square

Level 3 · basic

PUPP style · 3 points

g9.qe.3

g9.qe.4

The expression $x^2 - 6x + 11$ is given.

- a) Show that $x^2 - 6x + 11 = (x - 3)^2 + 2$.
- b) Solve the equation $x^2 - 6x + 11 = 6$.
- c) Explain why the equation $x^2 - 6x + 11 = 0$ has no real roots.

MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

- a) $(x - 3)^2 + 2 = x^2 - 6x + 9 + 2 = x^2 - 6x + 11$. 1 pt
- b) $(x - 3)^2 = 4$, so $x = 5$ or $x = 1$. 1 pt
- c) $(x - 3)^2 = -2$ is impossible, because a square is never negative (or: $(x - 3)^2 + 2 \geq 2 > 0$). 1 pt

WORKED SOLUTION

- a) $(x - 3)^2 + 2 = x^2 - 6x + 9 + 2 = x^2 - 6x + 11$.
- b) By a), the equation is $(x - 3)^2 + 2 = 6$, so $(x - 3)^2 = 4$. Then $x - 3 = 2$ or $x - 3 = -2$: $x = 5$ or $x = 1$.
Check: $25 - 30 + 11 = 6$ ✓ and $1 - 6 + 11 = 6$ ✓.
- c) By a), the equation is $(x - 3)^2 + 2 = 0$, that is, $(x - 3)^2 = -2$. A square is never negative, so no real x works. (Another way to say it: $(x - 3)^2 + 2$ is always at least 2, so it is never 0.)

Answer: b) $x = 5$ or $x = 1$; c) the equation becomes $(x - 3)^2 = -2$, and no square is negative.

TEACHER NOTES

For the teacher

Pupils can say: “I can complete the square in $x^2 + bx + c$, also when b is odd; I can solve a quadratic equation by completing the square, and tell from $(x + p)^2 = d$ whether it has two roots, one or none.”

Where this lesson sits. Grade 8 (31.2.1) completed the square “in simple cases” as an identity. Here it becomes a **method for solving** any quadratic equation, including odd b and $a \neq 1$. The hook runs lesson 1.1’s (D) → (E) backwards. The four moves are exactly the ones lesson 1.3 makes once with letters to prove the quadratic formula, so name them the same way in class.

Common mistakes and what helps

- Adding $(\frac{b}{2})^2$ to one side only (c2). Say “both sides” every time, and write the added number on both sides in the same colour.
- Not halving b , or squaring b itself (c1, c2, c3). Ask “what is the number in the bracket?” before squaring.
- Sign slips with negative b : half of -8 is -4 , but the number added is $(-4)^2 = 16$ (c1). The bracket keeps the sign: $(x - 4)^2$.
- Forgetting to divide c by a as well (Example 3, c7): $4x^2 - 8x + 3 = 0$ becomes $x^2 - 2x + \frac{3}{4} = 0$, not $x^2 - 2x + 3 = 0$.
- Fraction arithmetic: $-4 + \frac{25}{4} = \frac{9}{4}$. The Refresh box in Remember helps.
- In $(x + p)^2 + q$, adding $(\frac{b}{2})^2$ without taking it away again (c3).
- Forgetting the negative case after “take the root” (all Numeric items have a Why for this).

Key terms (EN • LT): completing the square • *dvinario kvadrato išskyrimas* (also *pilnojo kvadrato išskyrimas*); square of a binomial • *dvinario kvadratas*; plus or minus, $\pm$ • *plius minus*.

Timing (45 min). The lesson is tight for one period, so keep to the core:

- Remember: 4 min.
- Hook: 3 min.
- Explain: 10 min (the area model 3 min, the rule 3 min, the four moves 4 min). The frame “Two roots, one, or none” can be read quickly: it repeats lesson 1.1’s three cases.
- Examples 1–2 with their “Your turn”: 11 min. Example 3 ($a \neq 1$) with its “Your turn”: 5 min.
- Check c1–c5 (the core: both outcomes, including a rewrite as $(x + p)^2 + q$): 10 min.

c6 (put the steps in order), c7 (a full solution with $a \neq 1$) and the exam task are for pupils who finish early, or for homework. If the class is slow, take Example 3 and its “Your turn” as homework as well, and start the next lesson’s Remember with an $a \neq 1$ equation. The exam task continues lesson 1.1’s “hidden square”: now the left side is a square **plus** a number.

Programme: 32.1.1.1 “*Įrodoma ir taikoma kvadratinės lygties sprendinių formulė*” (this is the method the proof runs on), extending 31.2.1 “*Mokoma(si) paprastais atvejais iš kvadratinio trinario išskirti dvinario kvadratą.*” No PUPP item asks for completing the square directly; it underpins the formula (lesson 1.3) and the vertex of a parabola (spring). Achievements A1, A2.